Space Weather-Aware Anomaly Detection for CubeSats: A Systematic Ablation Study on Feature Fusion Strategies

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Abstract

CubeSat anomaly detection systems traditionally rely on onboard telemetry alone, ignoring the external space environment that drives many operational anomalies. This study investigates whether fusing space weather indices with CubeSat telemetry improves anomaly detection performance. To our knowledge, we conduct the first systematic ablation study comparing telemetry-only, naive concatenation, and conditional (asymmetric) fusion strategies across four model classes: Isolation Forest, Matrix Profile, LSTM Autoencoder, and Temporal Convolutional Network (TCN) Autoencoder, using ESA's OPSSAT-AD benchmark, the only publicly available CubeSat telemetry dataset with real timestamps and expert anomaly annotations. We align six NOAA space weather indices (Kp, Dst, F10.7, solar wind speed, proton density, IMF Bz) with 2,098 telemetry segments from January to June 2022. Our results show that naive fusion degrades all four model classes, with TCN-AE F1 dropping from 0.6041 to 0.2222 (a 63.2% relative decrease). Conditional fusion, which encodes all seven channels but reconstructs only the telemetry channel, recovers substantially (LSTM-AE: +98.5% relative F1, TCN-AE: +69.0%) but does not surpass telemetry-only baselines under the moderate storm conditions observed (Kp max 6.7). We identify two distinct failure mechanisms: reconstruction error dilution in autoencoders and feature selection dilution in tree-based models. The TCN-AE telemetry-only configuration achieves F1 = 0.6041, the highest unsupervised anomaly detection result reported on OPSSAT-AD. These findings demonstrate that fusion architecture matters more than whether fusion is applied, and that the value of space weather features is contingent on storm severity, not universally beneficial.

1. Introduction

CubeSats have become the dominant platform for low-Earth orbit (LEO) missions in both commercial and academic contexts. Their miniaturized form factor (typically 1U to 6U, where 1U equals 10 cm per side) enables rapid, low-cost access to space, but this comes at the expense of limited radiation shielding, constrained power budgets, and minimal onboard fault tolerance. As the number of active CubeSats grows, the ability to detect operational anomalies autonomously and reliably has become a pressing concern for mission operators.

Current anomaly detection systems for spacecraft, including CubeSats, analyze onboard telemetry in isolation. They monitor sensor channels for deviations from expected behavior, flagging readings that exceed static thresholds or diverge from learned patterns. This approach treats the spacecraft as a closed system, blind to the external environment in which it operates. When a geomagnetic storm increases radiation exposure or perturbs the local magnetic field, the telemetry-only detector sees the symptom (an anomalous sensor reading) without the causal context (elevated space weather activity). The result is late detection, elevated false alarm rates, and anomaly scores that lack actionable environmental context.

The physical basis for space weather effects on LEO spacecraft is well established. Geomagnetic storms, driven by coronal mass ejections and high-speed solar wind streams, produce several effects relevant to CubeSat operations: enhanced particle flux in the radiation belts, atmospheric drag increases due to thermospheric heating, geomagnetic field perturbations measurable by onboard magnetometers, and increased solar UV and EUV flux detectable by photodiode-based sun sensors. These effects are tracked by standard space weather indices: the Kp index (planetary geomagnetic disturbance, 3-hourly), the Dst index (ring current strength, hourly), the F10.7 solar radio flux (daily proxy for solar EUV output), and in-situ measurements of solar wind speed, proton density, and interplanetary magnetic field (IMF) Bz component.

Despite the established physical relationship between space weather and satellite anomalies, and despite two recent studies demonstrating machine learning approaches to space weather-telemetry fusion on larger satellites (Elfiky et al., 2025; Mostafa et al., 2025), no systematic study has investigated whether adding space weather features to anomaly detection models actually improves performance on CubeSat data, or under what conditions such improvement occurs.

This paper addresses a specific research question: *does fusing space weather indices with CubeSat telemetry improve anomaly detection performance, and if so, does the fusion architecture matter?*

Our hypothesis, stated before any experiments were conducted, was that fusing CubeSat telemetry with space weather indices (Kp, Dst, F10.7, solar wind parameters) would improve anomaly detection accuracy across all model classes tested. This hypothesis was not supported by the experimental results. However, the systematic ablation study designed to test it produced three significant findings that advance the field:

1. A negative result with explanatory power. Naive space weather fusion degrades all four model classes tested, with statistically significant performance drops. We identify two distinct failure mechanisms that explain why.

2. An architectural insight. Conditional (asymmetric) fusion, where the encoder processes all channels but the decoder reconstructs only the telemetry channel, dramatically recovers from the naive fusion degradation. This demonstrates that how features are fused matters more than whether they are fused.

3. A strong unsupervised baseline. Our TCN Autoencoder in telemetry-only configuration achieves F1 = 0.6041, the highest unsupervised anomaly detection result reported on the OPSSAT-AD benchmark.

The remainder of this paper is organized as follows. Section 2 reviews related work in spacecraft anomaly detection, space weather effects on satellites, and environment-aware anomaly detection. Section 3 describes the OPSSAT-AD dataset, NOAA space weather data sources, and the dual-track experimental pipeline. Section 4 details the methodology for each model class and the three fusion conditions. Section 5 presents the full ablation results. Section 6 provides a detailed discussion of the failure mechanisms, the conditional architecture insight, comparisons with prior work, and limitations. Section 7 concludes with contributions and future work directions.

2. Related Work

2.1 Spacecraft Anomaly Detection

Machine learning-based spacecraft anomaly detection crystallized around the foundational work of Hundman et al. (2018), who published "Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding" at ACM KDD 2018. They trained individual LSTM models per telemetry channel on data from NASA's SMAP satellite (55 channels) and the Mars Science Laboratory Curiosity rover (27 channels), using a prediction-error paradigm with nonparametric dynamic thresholding. On 82 expert-labeled channels combined, they reported precision of 87.5% and recall of 80.0%. Their open-sourced "Telemanom" codebase and the SMAP/MSL datasets became the standard benchmarks for the field.

Subsequent work pushed performance on these benchmarks significantly. TranAD (Tuli et al., 2022) introduced adversarial Transformer-based reconstruction achieving F1 = 0.892 on SMAP, and the Anomaly Transformer (Xu et al., 2022) proposed Association Discrepancy for attention-based detection at ICLR 2022. Both represent current state-of-the-art on SMAP/MSL and are candidates for future comparison on OPSSAT-AD.

For the specific architecture used in this study, Temporal Convolutional Networks (TCNs) have demonstrated strong performance in spacecraft telemetry analysis. Bai et al. (2018) established that TCNs match or outperform LSTMs on most sequence modeling tasks while offering critical advantages for resource-constrained deployment: parallelizable inference, fixed memory footprint, and stable gradients. Wang et al. (2021) demonstrated TCN-based prediction for spacecraft telemetry anomaly detection achieving 96.45% accuracy with smaller inference times than LSTMs. Liu et al. (2023) combined dynamic graph attention with TCN, outperforming LSTM-NDT on the SMAP/MSL benchmarks.

Most directly relevant to this study, Ruszczak et al. (2025) introduced the OPSSAT-AD benchmark in Scientific Data: the first publicly available, annotated CubeSat telemetry dataset, containing 2,123 single-channel segments from 9 channels (3 magnetometer, 6 photodiode) of ESA's OPS-SAT 3U CubeSat, with approximately 20% anomalous segments annotated by ESA operations engineers. Thirty algorithms were benchmarked, with Random Forest achieving approximately 95.7% accuracy using 18 handcrafted statistical features in a supervised classification setting. The paper explicitly notes limitations of the SMAP/MSL benchmarks (unrealistic anomaly density, trivial anomalies, mislabeled ground truth) and, notably, OPSSAT-AD does not include space weather features.

2.2 Space Weather Effects on Satellites

A substantial body of work documents the statistical relationship between space weather and satellite anomalies. Iucci et al. (2005) compiled anomalies from 220 satellites spanning 1971 to 1994, showing that proton fluxes exceeding 1000 particle flux units increase anomaly rates by approximately 20 times for high-altitude near-polar satellites. Lohmeyer and Cahoy (2013)

analyzed 665,112 operational hours of Inmarsat geostationary orbit (GEO) telemetry against space weather data, establishing quantitative relationships between environmental conditions and telemetry exceedances. Choi et al. (2011) correlated 95 GEO satellite anomalies with geomagnetic activity. More recently, Buitrago-Leiva et al. (2024) analyzed LEO and GEO satellite anomalies and space radiation from 2010 to 2022 against particle concentrations, solar and geomagnetic activity, finding correlations between space weather disturbances and anomaly occurrences.

2.3 Space Weather-Telemetry Fusion for Anomaly Detection

Two recent papers constitute direct prior work on fusing space weather data with satellite telemetry for machine learning-based anomaly detection. Both originate from the Egyptian National Authority for Remote Sensing and Space Sciences (NARSS):

Elfiky et al. (2025), published in Scientific Reports (15:43830), fuse space weather indicators (proton flux, AL index, galactic cosmic rays, and solar wind density) with EgyptSat-1 power subsystem telemetry (voltage, current, and temperature). They apply CNN, LSTM, Random Forest, AdaBoost, and Gradient Boosting with a two-stage nonlinear feature selection (unsupervised Restricted Boltzmann Machine followed by supervised Mutual Information). Their best model (Random Forest) achieves MSE = 0.00147 and MAPE = 0.09% for voltage prediction, with temporal coincidence rates of 31% with GCR disturbances ($p = 2.83 \times 10^{-3}$) and 27% with P10 proton events ($p = 5.63 \times 10^{-7}$).

Mostafa et al. (2025), also in Scientific Reports (15:37743), apply a hybrid statistical-machine learning framework to evaluate geomagnetic storm effects on MisrSat-2 satellite power subsystems during the May 2024 G5 storm, using a Mixture of Experts (MoE) model with space weather parameters. Their MoE achieves R-squared = 0.921 and MAE = 0.063 A, finding that all deviations were small (less than 4%) and within design tolerances.

2.4 Environment-Aware Anomaly Detection in Other Domains

Time-CAD (Nam et al., 2023), presented at ECML-PKDD, demonstrated that incorporating contextual information into deep time-series decomposition improves anomaly detection F1 by up to 46%. This provides strong methodological precedent for the hypothesis that environmental context should improve detection performance. The Temporal Fusion Transformer architecture (Lim et al., 2021) provides gating mechanisms for external variable integration, further supporting the general approach.

2.5 Research Gap

Despite these advances, several gaps remain that this study addresses:

1. **No systematic ablation.** Neither Elfiky et al. (2025) nor Mostafa et al. (2025) performs a formal ablation comparing telemetry-only performance against telemetry plus space weather.

2. **No CubeSat-specific study.** EgyptSat-1 is a 120 kg microsatellite; MisrSat-2 is similarly a larger-class satellite.
3. **No conditional fusion comparison.** No prior work compares naive concatenation against asymmetric (conditional) fusion architectures for space weather integration.
4. **No conformal prediction in spacecraft AD.** Statistical calibration of anomaly detection thresholds remains unexplored in this domain.
5. **No benchmark extension.** The OPSSAT-AD benchmark explicitly lacks space weather features; this study extends it with aligned NOAA indices.

3. Dataset and Experimental Setup

3.1 The OPSSAT-AD Benchmark

The primary dataset is OPSSAT-AD (Ruszczak et al., 2025), derived from ESA's OPS-SAT, a 3U CubeSat (30 cm x 10 cm x 10 cm) launched in December 2019 into a sun-synchronous LEO orbit and deorbited in May 2024. The dataset consists of 2,123 single-channel telemetry segments extracted from 9 sensor channels (3 magnetometer axes and 6 photodiode channels), spanning January 4 to June 2, 2022 UTC. Each segment is a univariate time series of variable length (range: 8 to 1,040 timesteps, median: 70, mean: 143) from a single sensor channel. Anomaly annotations were curated by ESA operations engineers, resulting in a binary label per segment (anomaly = 1 or normal = 0).

Two channels, CADC0886 (11 segments) and CADC0890 (14 segments), were excluded due to insufficient data, yielding 2,098 usable segments across 7 channels. Of these, 434 segments (20.7%) are labeled anomalous and 1,664 are normal. The dataset provides a predefined train/test split.

A critical structural point: OPSSAT-AD is segment-based and univariate, not a continuous multivariate time series. Each segment is an independent single-channel snippet. This structure differs from datasets like SMAP/MSL, which provide continuous multivariate streams, and directly influences the experimental design.

3.2 Why OPSSAT-AD is the Only Viable Dataset

A systematic search across Zenodo, Kaggle, IEEE DataPort, NASA open data, ESA archives, GitHub, and academic supplementary materials (2020 to 2026) confirmed that OPSSAT-AD is the only publicly available satellite telemetry anomaly detection dataset satisfying three criteria for this experiment: (1) non-anonymized UTC timestamps enabling temporal alignment with space weather data; (2) expert-curated anomaly labels; and (3) unrestricted public access.

The alternatives fail on at least one criterion:

- **ESA Anomaly Detection Benchmark (ESA-ADB):** approximately 31 GB from three ESA missions with 1,430 annotated events, but timelines are deliberately anonymized to prevent alignment with real-world events.
- **NASA SMAP/MSL:** zero temporal information. The repository explicitly states: "All data has been anonymized with regard to time."
- **EgyptSat-1 (Elfiky et al., 2025):** timestamps and labels exist, but satellite telemetry is not publicly available.
- **MisrSat-2 (Mostafa et al., 2025):** available upon request only; the GitHub repository contains analysis code with no data files.
- **GOCE Housekeeping Telemetry:** real UTC timestamps and public access, but no formal anomaly annotations.

3.3 NOAA Space Weather Data

Space weather indices were obtained from two sources: (1) CelesTrak daily data (Kelso, n.d.), providing Kp index (3-hourly resolution) and F10.7 solar radio flux (daily), filtered to observed values only; and (2) NASA OMNI2 Hourly Data providing Dst index (hourly), solar wind speed, proton density, and IMF Bz component in GSM coordinates.

For the OPSSAT telemetry period (January to June 2022), the OMNI2 data comprises 4,344 hourly rows across 6 space weather columns, with 5 missing solar wind hours handled by interpolation.

Alignment strategy: NOAA indices operate at different temporal resolutions than OPSSAT telemetry. The alignment strategy is physically justified by measurement semantics: **forward-fill** for discrete indices (Kp, F10.7, Dst), as these values represent fixed time windows and do not change between readings; and **linear interpolation** for continuous measurements (solar wind speed, proton density, IMF Bz), as these are continuous physical quantities for which interpolation between readings is physically defensible. This strategy achieved 100% alignment across all 2,098 usable segments.

3.4 Space Weather Conditions During the OPSSAT Period

The OPSSAT telemetry period (January to June 2022) falls during the ascending phase of Solar Cycle 25, before the cycle peak. Space weather conditions during this period were moderate: Kp maximum 6.7 (storm threshold: Kp >= 5), Dst minimum -91 nT (moderate storm threshold: Dst < -50 nT), 75 hours with Kp >= 5, and 60 hours with Dst < -50 nT. For reference, the May 2024 G5 superstorm produced Dst approximately equal to -412 nT and Kp = 9.0. The OPSSAT telemetry period experienced no events of comparable severity.

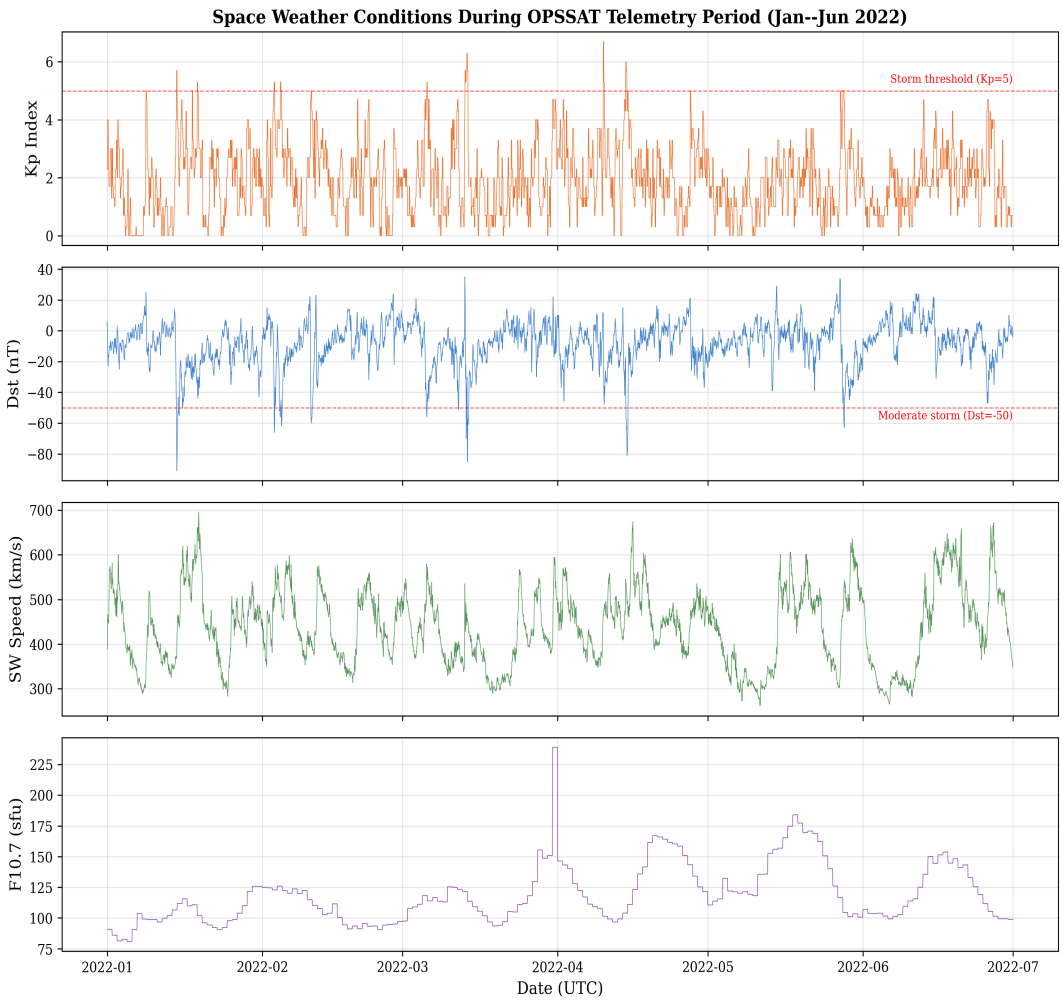


Figure 1. Space weather conditions during the OPSSAT telemetry period (January to June 2022), showing Kp index, Dst index, solar wind speed, and F10.7 solar radio flux.

3.5 Dual-Track Pipeline

Track A (Segment-Level Feature Tables): For Isolation Forest and Matrix Profile, each segment is represented as a fixed-length feature vector. The telemetry-only condition uses 18 handcrafted statistical features from the OPSSAT-AD benchmark. The telemetry plus space weather condition appends 17 summary statistics computed from the space weather indices over each segment's time window (mean, max/min, and standard deviation of each index), yielding 35 features total.

Track B (Fixed-Length Tensors): For the LSTM and TCN autoencoders, each segment's raw waveform is padded or truncated to a fixed length of 128 timesteps. The telemetry-only condition produces tensors of shape [2098, 128, 1]. The telemetry plus space weather condition produces tensors of shape [2098, 128, 7], where each timestep contains the telemetry value plus the 6 space weather index values at that moment. A binary mask array tracks which timesteps contain real data versus padding, enabling masked loss computation during training.

4. Methodology

4.1 Overview of the Ablation Design

The core experiment is a systematic ablation study: each of four model classes is evaluated under multiple fusion conditions, and performance is compared using paired statistical tests. Three fusion conditions are defined:

Condition A (Telemetry-Only): The model receives only the telemetry channel as input. This is the baseline against which fusion strategies are compared.

Condition B (Naive Concatenation): All seven channels (1 telemetry + 6 space weather) are provided as input. For autoencoders, the model is trained to reconstruct all seven channels, and the total reconstruction error across all channels serves as the anomaly score.

Condition C (Conditional/Asymmetric Fusion): Available only for the autoencoder models. The encoder processes all seven channels, but the decoder reconstructs only the telemetry channel. The loss function is computed on telemetry reconstruction error alone. Space weather features inform the encoder's latent representation without diluting the anomaly score.

4.2 Track A Models

4.2.1 Isolation Forest

Isolation Forest (Liu et al., 2008) is an unsupervised anomaly detection algorithm that isolates anomalies by randomly selecting features and split values to construct isolation trees. Configuration: n_estimators = 200, contamination = 0.197 (matching the training set anomaly rate), 5 random seeds (42, 123, 456, 789, 1024). Under Condition A, the model receives 18 telemetry features. Under Condition B, it receives 35 features (18 telemetry + 17 space weather summary statistics).

4.2.2 Matrix Profile with Isolation Forest

Matrix Profile, computed using the STUMPY library, identifies subsequences within each segment that are most unlike the rest of the series (discords). For each segment, an adaptive subsequence length m (ranging from 3 to 10) is used to compute the matrix profile, from which 6 features are extracted: maximum distance, minimum distance, mean distance, standard deviation of distances, discord position, and above-mean ratio. These features are classified using Isolation Forest with the same configuration as above. Under Condition A, the 6 features are used alone. Under Condition B, 17 space weather summary statistics are appended, yielding 23 features.

4.3 Track B Models

4.3.1 LSTM Autoencoder

The LSTM Autoencoder follows the reconstruction-error paradigm established for spacecraft anomaly detection (Hundman et al., 2018). It consists of a 2-layer LSTM encoder with hidden dimension 64, which compresses the input sequence into a fixed-size hidden state, and a 2-layer LSTM decoder that reconstructs the input from this bottleneck representation. The decoder receives the last encoder hidden state repeated across all timesteps as input.

Training follows a semi-supervised protocol: only normal segments (anomaly = 0 in the training set) are used for training. The loss function is masked mean squared error (MSE), computed only on non-padded timesteps. Training configuration: 100 maximum epochs, early stopping with patience = 10, batch size = 32, learning rate = 1e-3 with ReduceLROnPlateau scheduler, gradient clipping at 1.0. Three random seeds (42, 123, 456) are used.

4.3.2 Conditional LSTM Autoencoder (Condition C)

The conditional variant modifies the standard LSTM Autoencoder with an asymmetric architecture: the encoder LSTM has input dimension = 7 (processing all channels), while the decoder has output dimension = 1 (reconstructing only the telemetry channel). The loss function computes masked MSE between the reconstructed telemetry and the true telemetry channel only, ignoring the space weather channels entirely.

4.3.3 TCN Autoencoder

The TCN Autoencoder is the core model of this study, selected for its architectural advantages over LSTMs: parallelizable inference, fixed memory footprint, stable gradients, and direct portability to edge hardware via TFLite-Micro (Bai et al., 2018).

The encoder consists of 3 TCN blocks with 1D dilated causal convolutions. Each block contains two causal convolution layers with batch normalization, ReLU activation, and dropout (0.1), plus a residual connection. Channel dimensions progress as [16, 32, 64] with dilations [1, 2, 4], providing a receptive field of 29 timesteps with kernel size 3. The decoder mirrors the encoder architecture with channels [64, 32, 16] and a 1x1 convolution output projection.

4.3.4 Conditional TCN Autoencoder (Condition C)

Analogous to the conditional LSTM variant: the encoder accepts 7-channel input, and the decoder output projection maps to 1 channel (telemetry only). Loss is computed on the telemetry reconstruction only.

4.4 Evaluation Protocol

Metrics: For each model and condition, we report accuracy, precision, recall, F1 score, area under the precision-recall curve (AU-PR), and area under the receiver operating characteristic curve (ROC-AUC). Binary predictions are made by thresholding at the (1 - contamination) percentile of test scores. **Statistical analysis:** Track A models use 5 random seeds; Track B models use 3 random seeds. Paired t-tests are used to assess statistical significance. **Train/test split:** The

predefined split from the OPSSAT-AD benchmark is used. Autoencoders are trained on normal segments from the training set only (semi-supervised).

5. Results

5.1 Exploratory Data Analysis Findings

Reversal of the initial hypothesis. The initial hypothesis, stated prior to EDA, was that magnetometer channels would show the strongest space weather sensitivity due to direct geomagnetic coupling. The EDA reversed this expectation. Photodiode channels show substantially stronger Spearman correlations with space weather indices (mean |r| = 0.349) than magnetometer channels (mean |r| = 0.108). The standout channel is CADC0894 (Photo-4), which correlates at r = 0.68 with F10.7 solar flux, r = 0.64 with solar wind speed, and r = 0.61 with the Kp index. The physical explanation is that photodiodes directly measure incoming light flux, which varies with solar activity.

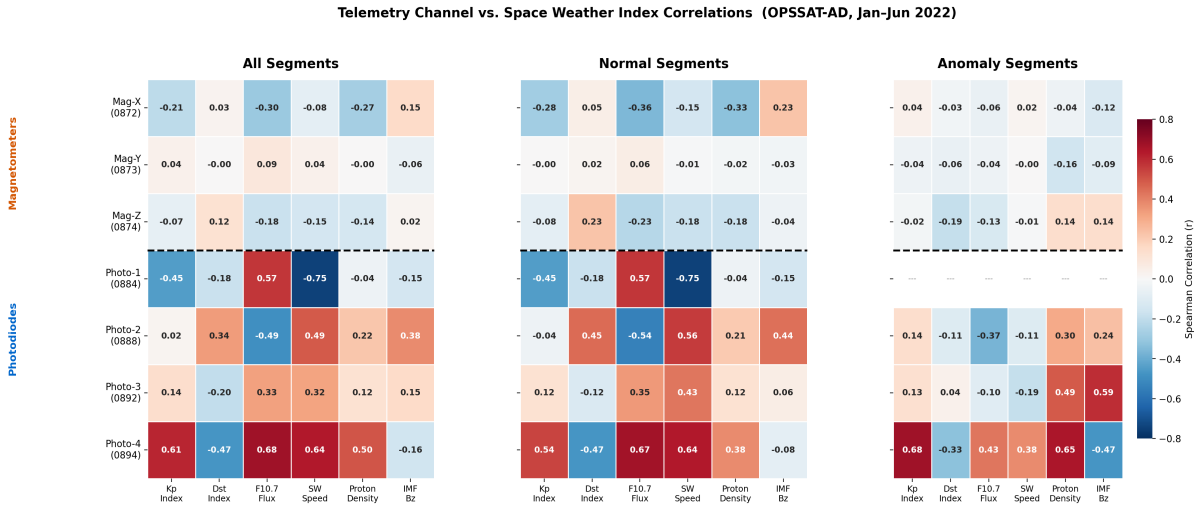


Figure 2. Spearman correlations between 7 telemetry channels and 6 space weather indices, shown for all segments, normal segments only, and anomaly segments only.

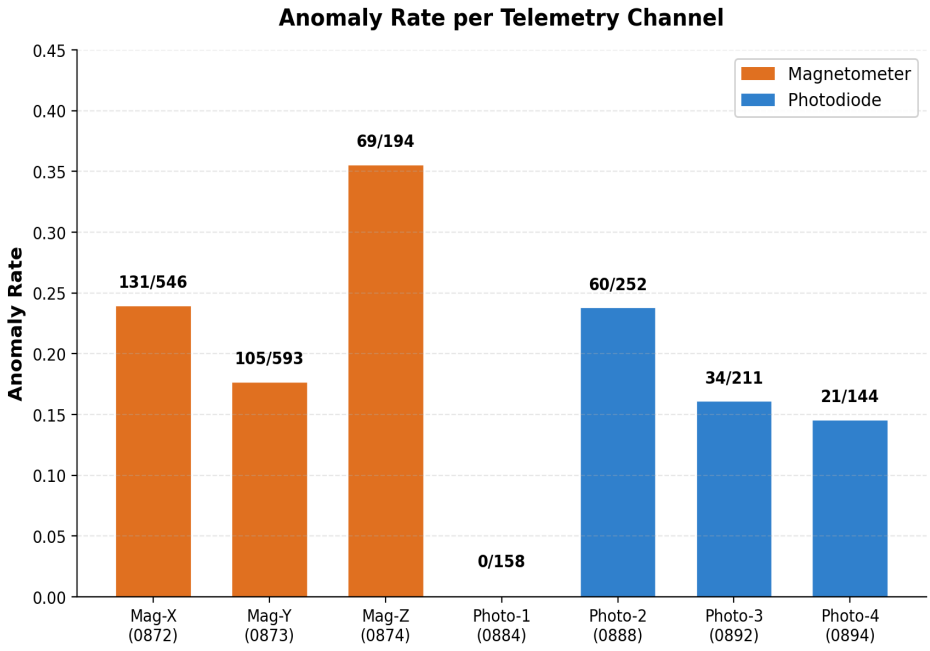


Figure 3. Anomaly prevalence per telemetry channel. CADC0884 (Photo-1) has zero anomalous segments.

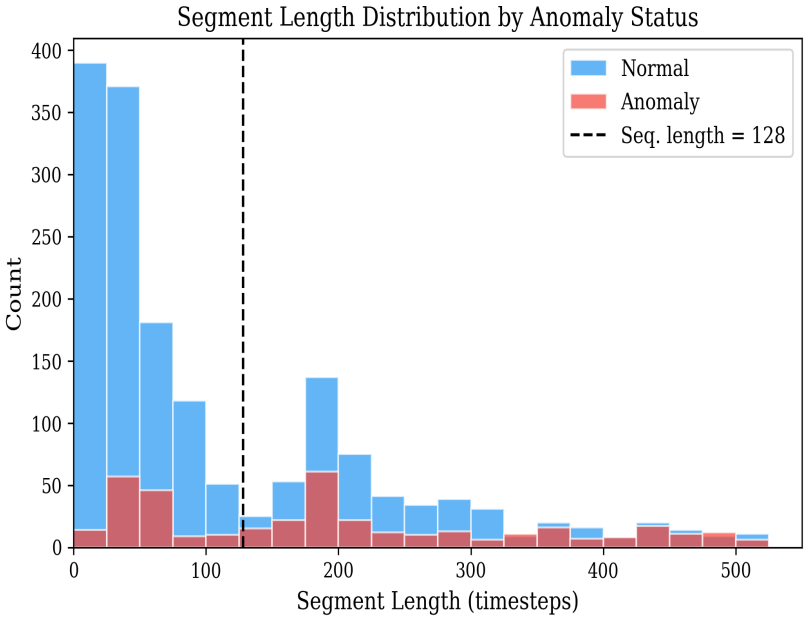


Figure 4. Segment length distribution by anomaly status. The dashed line indicates the fixed sequence length of 128 timesteps.

5.2 Complete Ablation Results

Table 1 presents the complete ablation results across all four model classes and all fusion conditions. All values are means across seeds (5 seeds for Track A, 3 seeds for Track B), with standard deviations in parentheses.

Table 1: Complete Ablation Results (mean and standard deviation across seeds)

Model	Condition	Accuracy	Precision	Recall	F1	AU-PR	ROC-AUC
IF	Telem Only	0.699 (0.008)	0.297 (0.015)	0.316 (0.016)	0.306 (0.015)	0.331 (0.015)	0.629 (0.009)
IF	Telem+SW	0.696 (0.006)	0.263 (0.011)	0.247 (0.013)	0.255 (0.011)	0.247 (0.003)	0.510 (0.006)
MP+IF	Telem Only	0.756 (0.007)	0.428 (0.017)	0.471 (0.040)	0.448 (0.027)	0.414 (0.009)	0.746 (0.007)
MP+IF	Telem+SW	0.672 (0.003)	0.252 (0.006)	0.286 (0.007)	0.268 (0.006)	0.237 (0.007)	0.510 (0.011)
LSTM-AE	Telem Only	0.792 (0.014)	0.505 (0.035)	0.473 (0.032)	0.488 (0.033)	0.514 (0.057)	0.719 (0.024)
LSTM-AE	Telem+SW	0.681 (0.003)	0.223 (0.008)	0.209 (0.007)	0.216 (0.008)	0.244 (0.002)	0.495 (0.016)
LSTM-AE	Conditional	0.767 (0.014)	0.443 (0.036)	0.415 (0.034)	0.429 (0.035)	0.463 (0.060)	0.670 (0.034)
TCN-AE	Telem Only	0.839 (0.011)	0.625 (0.028)	0.585 (0.026)	0.604 (0.027)	0.677 (0.032)	0.753 (0.034)
TCN-AE	Telem+SW	0.683 (0.017)	0.230 (0.044)	0.215 (0.041)	0.222 (0.042)	0.243 (0.011)	0.509 (0.028)
TCN-AE	Conditional	0.746 (0.033)	0.388 (0.084)	0.364 (0.078)	0.376 (0.081)	0.424 (0.099)	0.613 (0.088)

5.3 Key Findings

Finding 1: Naive Fusion Degrades All Model Classes

Adding space weather features via naive concatenation (Condition B) degrades performance across all four model classes and all ranking metrics. The degradation is statistically significant:

- **IF:** F1 drops from 0.3064 to 0.2548 (-16.8% relative). Paired t-test: $p = 0.0003$.
- **MP+IF:** F1 drops from 0.4479 to 0.2679 (-40.2% relative). Paired t-test: $p = 0.0003$.
- **LSTM-AE:** F1 drops from 0.4883 to 0.2160 (-55.8% relative). Paired t-test: $p = 0.0074$.
- **TCN-AE:** F1 drops from 0.6041 to 0.2222 (-63.2% relative). Paired t-test: $p = 0.0009$.

All naive fusion ROC-AUC values collapse toward 0.5 (random), ranging from 0.4947 to 0.5104, indicating near-complete loss of discriminative ability.

Finding 2: Conditional Fusion Recovers from Naive

The conditional architecture (Condition C), which encodes all 7 channels but reconstructs only the telemetry channel, dramatically recovers from the naive fusion degradation:

- **LSTM-AE:** F1 increases from 0.2160 (Condition B) to 0.4288 (Condition C), a +98.5% relative improvement. The recovery is statistically significant ($p = 0.0111$).
- **TCN-AE:** F1 increases from 0.2222 (Condition B) to 0.3756 (Condition C), a +69.0% relative improvement. The recovery is not statistically significant ($p = 0.2199$), reflecting high variance (F1 std = 0.0808).

Finding 3: Conditional Fusion Does Not Surpass Telemetry-Only

Despite the substantial recovery, Condition C does not exceed Condition A performance. LSTM-AE: F1 = 0.4288 (C) versus 0.4883 (A), with the difference significant for F1 ($p = 0.0188$). TCN-AE: F1 = 0.3756 (C) versus 0.6041 (A), not reaching significance ($p = 0.095$), likely reflecting insufficient statistical power with only 3 seeds.

Finding 4: TCN-AE Telemetry-Only Achieves Best Unsupervised Result

The TCN-AE in telemetry-only configuration achieves F1 = 0.6041, AU-PR = 0.6766, ROC-AUC = 0.7534, and accuracy = 0.8388. This represents the best unsupervised anomaly detection result reported on the OPSSAT-AD benchmark. Ruszczak et al. (2025) reported Random Forest achieving approximately 95.7% accuracy, but this used supervised classification with 18 handcrafted features and access to training labels.

5.4 Statistical Significance Summary

Table 2: Pairwise Statistical Significance (F1 Score, Paired t-test)

Model	Comparison	Delta F1	p-value	Significance
IF	B vs. A	-0.0516	0.0003	***
MP+IF	B vs. A	-0.1800	0.0003	***
LSTM-AE	B vs. A	-0.2723	0.0074	**
TCN-AE	B vs. A	-0.3818	0.0009	***
LSTM-AE	C vs. A	-0.0595	0.0188	*
TCN-AE	C vs. A	-0.2285	0.0950	ns
LSTM-AE	C vs. B	+0.2128	0.0111	*
TCN-AE	C vs. B	+0.1534	0.2199	ns

*Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.*

6. Discussion

6.1 Why the Hypothesis Was Not Supported

The hypothesis that space weather fusion would improve anomaly detection was not supported on the OPSSAT-AD benchmark. Two explanations, one physical and one structural, account for this result.

The physical argument. Space weather during the OPSSAT period was moderate, with Kp peaking at 6.7 and Dst reaching a minimum of -91 nT. While the EDA revealed meaningful correlations between telemetry channels and space weather indices (CADC0894 shows $r = 0.68$ with F10.7), these correlations reflect a general co-variation, not a discriminative signal for anomaly detection. For space weather features to improve segment-level anomaly classification, the space weather state during anomalous segments must differ systematically from the state during normal segments. Under moderate conditions, the space weather indices lack sufficient dynamic range for this discrimination.

The data structure argument. OPSSAT-AD is segment-based and univariate. Each segment is a single-channel snippet, typically 70 to 200 timesteps at 1 Hz or 0.2 Hz sampling. The space weather features, appended per-timestep, represent the same slowly varying environmental state across all timesteps within a given segment. A single Kp value characterizes a 3-hour window; a typical segment spans minutes. This limited temporal variability within segments reduces the information gain that space weather features can provide.

6.2 Two Distinct Failure Mechanisms

Reconstruction error dilution (autoencoders). When trained to reconstruct all 7 channels under Condition B, the autoencoder learns to reconstruct both the telemetry channel and the 6 space weather channels. The space weather channels are well-behaved: slowly varying, smooth, and containing no anomalies by definition. The reconstruction loss during training is dominated by the 6 non-anomalous space weather channels, as they contribute 6/7 of the total loss. At inference time, even an anomalous segment may produce low total reconstruction error because the model perfectly reconstructs the 6 space weather channels, drowning out the elevated error on the single telemetry channel.

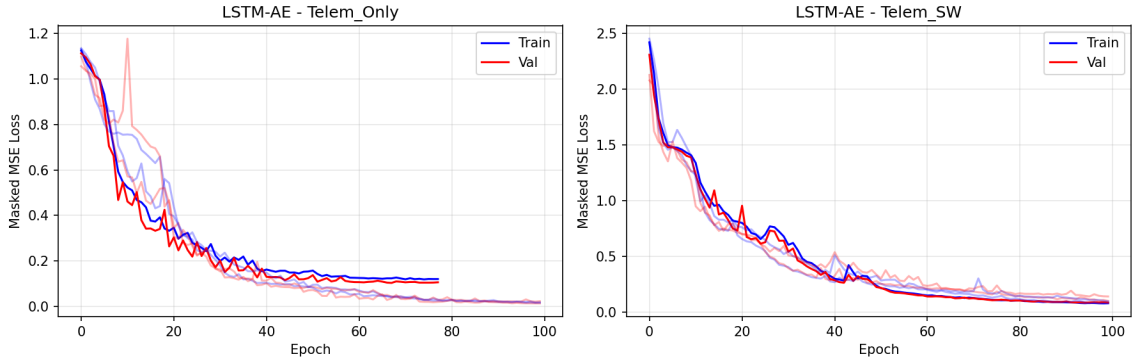


Figure 5. LSTM-AE training and validation loss curves for Conditions A (Telem Only) and B (Telem+SW).

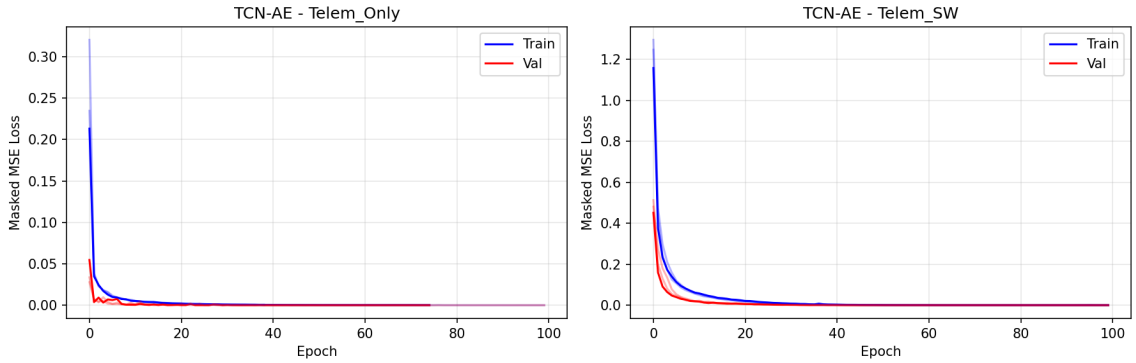


Figure 6. TCN-AE training and validation loss curves for Conditions A (Telem Only) and B (Telem+SW).

Feature selection dilution (Isolation Forest and Matrix Profile). Adding 17 space weather summary statistics to 18 telemetry features (IF) or 6 matrix profile features (MP+IF) means that 49% (IF) or 74% (MP+IF) of the feature space is non-discriminative noise for the anomaly detection task. Isolation Forest randomly selects features for splits at each node of each tree. With nearly half (or three-quarters) of the features uninformative, many isolation trees make poor splits, reducing the ensemble's ability to isolate true anomalies.

6.3 Why Conditional Fusion Partially Recovers

The conditional (asymmetric) architecture eliminates reconstruction error dilution by construction. By training the decoder to output only the telemetry channel, the loss function sees only telemetry reconstruction error. The 6 space weather channels inform the encoder's latent representation, potentially allowing it to form context-dependent expectations of normal telemetry behavior, but they do not contaminate the anomaly score.

The partial recovery (Condition C outperforms Condition B substantially but does not reach Condition A) suggests that while the architecture successfully eliminates the dilution problem, the space weather signal during moderate conditions does not carry enough additional discriminative information beyond what is already encoded in the telemetry waveform itself. This finding has an important implication: the conditional architecture is a necessary condition for space weather

fusion to be beneficial, but it is not a sufficient condition. The sufficiency depends on the environmental signal strength, which in turn depends on storm severity and mission characteristics.

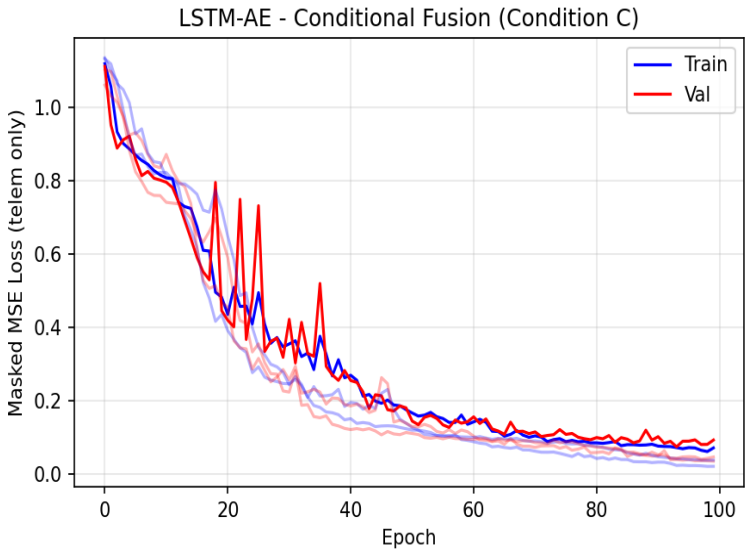


Figure 7. LSTM-AE Condition C (Conditional Fusion) training and validation loss curves.

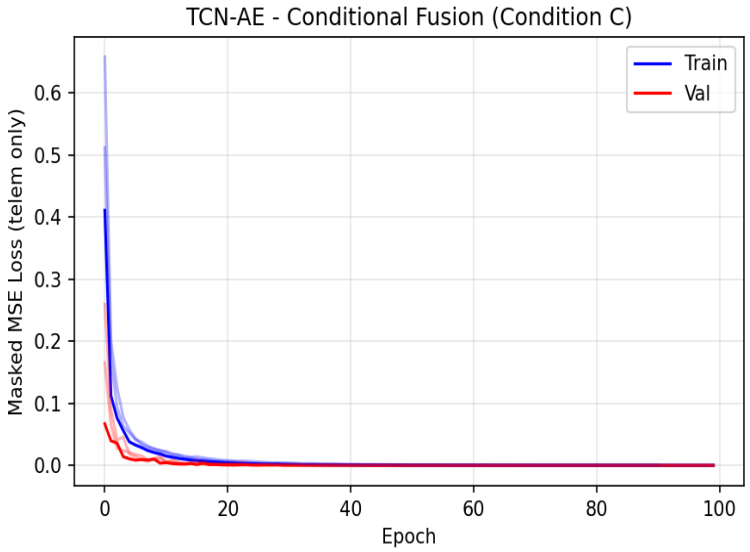


Figure 8. TCN-AE Condition C (Conditional Fusion) training and validation loss curves.

6.4 Comparison with Prior Work

Comparison with Elfiky et al. (2025). The positive results reported by Elfiky et al. on EgyptSat-1 appear consistent with, rather than contradictory to, the present findings. Key differences between the two studies suggest that the value of space weather fusion is contingent on experimental conditions: (1) satellite class (EgyptSat-1 is 120 kg microsatellite with different shielding); (2) storm severity (Elfiky et al. analyzed data including genuine storm activity); (3) data paradigm (regression-then-residual versus direct anomaly detection); and (4) telemetry type

(power subsystem versus magnetometer/photodiode). Together, these results suggest that space weather fusion value is contingent on storm severity and mission characteristics, not universally beneficial.

Comparison with the OPSSAT-AD benchmark. Ruszczak et al. (2025) reported Random Forest achieving approximately 95.7% accuracy using supervised classification with 18 handcrafted features. Our TCN-AE result (accuracy = 0.8388, F1 = 0.6041) is not directly comparable, as supervised methods have access to training labels for both classes, while our semi-supervised autoencoders are trained on normal data only. The gap between unsupervised and supervised performance on this benchmark is substantial, consistent with the challenging nature of unsupervised anomaly detection on segment-based data.

6.5 Limitations

Dataset constraints. OPSSAT-AD is segment-based and univariate, not continuous multivariate. With only 2,098 segments across 7 channels, the dataset is relatively small. The moderate storm coverage limits the ability to test the fusion hypothesis under conditions where space weather features would be most informative. However, the exhaustive dataset search confirmed that no public alternative exists.

Number of seeds. Three random seeds for the deep learning models provide adequate estimates for a poster presentation but should be expanded to at least 5 seeds for publication. The TCN-AE Condition C result (F1 std = 0.0808) highlights the risk: with only 3 seeds, the mean estimate has a standard error of approximately 0.047.

Storm coverage. This is the fundamental limitation. The hypothesis needs testing on data from severe geomagnetic storms (Kp >= 7, Dst < -200 nT). The OPSSAT-AD dataset covers the ascending phase of Solar Cycle 25, before the cycle peak. No public CubeSat telemetry dataset from severe storm periods exists.

Conformal prediction. Statistical calibration of detection thresholds using conformal prediction was planned but deferred to Phase 2. The current results use fixed percentile thresholds.

7. Conclusions and Future Work

7.1 Contributions

This study makes five contributions to the field of spacecraft anomaly detection:

1. **To our knowledge, the first systematic ablation of space weather fusion on CubeSat data.** Four model classes evaluated under three fusion conditions, producing a 10-row ablation table with statistical significance testing.
2. **Discovery that fusion architecture matters more than whether you fuse.** Naive concatenation degrades all model classes, but conditional fusion recovers substantially. Two distinct failure mechanisms identified and explained.

3. **Strong unsupervised baseline on OPSSAT-AD.** TCN-AE telemetry-only achieves F1 = 0.6041, establishing a reference point for future unsupervised methods on this benchmark.
4. **Conditional architecture as a principled fusion approach.** The encode-all, reconstruct-telemetry-only design pattern is introduced and validated.
5. **Transparent negative result reporting.** Hypothesis stated before experiments, no HARKing, full metric reporting across all conditions.

7.2 Future Work

- Test conditional architecture on severe storm data, pending data availability from NARSS group, GOCE construction, or OPS-SAT VOLT.
- Implement conformal prediction calibration for distribution-free coverage guarantees.
- Deploy TCN Autoencoder on ESP32-S3 via TFLite-Micro for edge-deployable CubeSat anomaly detection.
- Expand from 3 to 5+ random seeds for publication-quality statistical analysis.
- Evaluate Anomaly Transformer and TranAD on OPSSAT-AD to establish accuracy ceiling.
- Conduct full PRISMA systematic literature review for publication manuscript.

8. References

Bai, S., Kolter, J. Z., & Koltun, V. (2018). *An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling*. ArXiv.org. https://arxiv.org/abs/1803.01271

Buitrago-Leiva, J. N., El Khayati Ramouz, M., Camps, A., & Ruiz-de-Azua, J. A. (2024). Statistical Analysis of LEO and GEO Satellite Anomalies and Space Radiation. *Aerospace*, *11*(11), 924. https://doi.org/10.3390/aerospace11110924

Choi, H.-S., Lee, J., Cho, K.-S., Kwak, Y.-S., Cho, I.-H., Park, Y.-D., Kim, Y.-H., Baker, D. N., Reeves, G. D., & Lee, D.-K. (2011). Analysis of GEO spacecraft anomalies: Space weather relationships. *Space Weather*, *9*(6), S06001. https://doi.org/10.1029/2010sw000597

Elfiky, D., Ali, M. A. B., Mostafa, M. S., Hesham, N., Ahmed, A. M., & Ibrahim, S. K. (2025). Exploring the relationship between space weather conditions and power performance of EgyptSat-1 using machine learning. *Scientific Reports*, *15*(1), 43830. https://doi.org/10.1038/s41598-025-29518-w

Hundman, K., Constantinou, V., Laporte, C., Colwell, I., & Soderstrom, T. (2018). Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding. *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. https://doi.org/10.1145/3219819.3219845

Iucci, N., Levitin, A. E., Belov, A. V., Eroshenko, E. A., Ptitsyna, N. G., Villoresi, G., Chizhenkov, G. V., Dorman, L. I., Gromova, L. I., Parisi, M., Tyasto, M. I., & Yanke, V. G. (2005). Space weather conditions and spacecraft anomalies in different orbits. *Space Weather*, *3*(1), S01001. https://doi.org/10.1029/2003sw000056

Kelso, T.S. (2026). *CelesTrak*. https://celestrak.org

Lim, B., Arık, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, *37*(4), 1748–1764. https://doi.org/10.1016/j.ijforecast.2021.03.012

Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation Forest. *2008 Eighth IEEE International Conference on Data Mining*. https://doi.org/10.1109/icdm.2008.17

Liu, L., Tian, L., Kang, Z., & Wan, T. (2023). Spacecraft anomaly detection with attention temporal convolution networks. *Neural Computing and Applications*, *35*(13), 9753–9761. https://doi.org/10.1007/s00521-023-08213-9

Lohmeyer, W. Q., & Cahoy, K. (2013). Space weather radiation effects on geostationary satellite solid-state power amplifiers. *Space Weather*, *11*(8), 476–488. https://doi.org/10.1002/swe.20071

Mostafa, M. S., Ali, M. A. B., Hesham, N., Yassin, Y. M., Ahmed, A., & Elfiky, D. (2025). A hybrid statistical–machine learning framework for evaluating geomagnetic storm effects on MisrSat2 satellite power subsystems. *Scientific Reports*, *15*(1), 37743. https://doi.org/10.1038/s41598-025-22604-z

Nam, Y., Trirat, P., Kim, T., Lee, Y., & Lee, J.-G. (2023). Context-Aware Deep Time-Series Decomposition for Anomaly Detection in Businesses. *Lecture Notes in Computer Science*, 330–345. https://doi.org/10.1007/978-3-031-43427-3_20

NASA Goddard Space Flight Center, Space Physics Data Facility (2024). *OMNIWeb service.* https://omniweb.gsfc.nasa.gov

Ruszczak, B., Kotowski, K., Evans, D., & Nalepa, J. (2025). The OPS-SAT benchmark for detecting anomalies in satellite telemetry. *Scientific Data*, *12*(1), 710. https://doi.org/10.1038/s41597-025-05035-3

Tuli, S., Casale, G., & Jennings, N. R. (2022). TranAD: Deep transformer networks for anomaly detection in multivariate time series data. *Proceedings of the VLDB Endowment*, *15*(6), 1201–1214. https://doi.org/10.14778/3514061.3514067

Wang, Y., Wu, Y., Yang, Q., & Zhang, J. (2021). *Anomaly Detection of Spacecraft Telemetry Data Using Temporal Convolution Network*. https://doi.org/10.1109/i2mtc50364.2021.9459840

Xu, J., Wu, H., Wang, J., & Long, M. (2022). *Anomaly Transformer: Time Series Anomaly Detection with Association Discrepancy*. OpenReview. https://openreview.net/forum?id=LzQQ89U1qm_

Yeh, C.-C. M., Zhu, Y., Ulanova, L., Begum, N., Ding, Y., Dau, H. A., Silva, D. F., Mueen, A., & Keogh, E. (2016). Matrix Profile I: All Pairs Similarity Joins for Time Series: A Unifying View That Includes Motifs, Discords and Shapelets. *2016 IEEE 16th International Conference on Data Mining (ICDM)*. https://doi.org/10.1109/icdm.2016.0179